SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 2.6: Surface Area of Solids — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.6: Surface Area of Solids
Driving Question	DRIVING QUESTION: How do we calculate exactly how much material is needed to cover or construct any solid shape — and why does the shape of a container matter even when the volume stays the same?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Nets of Cones: How Much Metal Makes a Bucket?**	Students examine a real or illustrated metal bucket (cone-shaped, as commonly used by mjenzi workers and street vendors in Nairobi) and attempt to estimate how much sheet metal was used to make it. When the bucket's lateral surface is unrolled (physically or via diagram), it forms a flat sector of a circle — not a rectangle. Teacher should display an actual net cut from cardboard and ask students to measure its parts before any formula is given. Note: students typically underestimate because they ignore the circular base.	The lateral (curved) surface of a cone unfolds into a sector of a circle with radius equal to the slant height l and arc length equal to the circumference of the base ($2\pi r$). Therefore: Lateral Surface Area = πrl. Adding the circular base gives Total Surface Area of a cone: TSA = $\pi rl + \pi r^2$, where r = base radius and l = slant height. Slant height is related to perpendicular height h by: $l = \sqrt{r^2 + h^2}$. Pedagogical note — insist students distinguish between h (vertical height) and l (slant height) from lesson 1 onward; this confusion is the single most common error across all frustum lessons.	Students apply TSA = $\pi rl + \pi r^2$ to calculate the sheet metal needed for a cone-shaped water bucket with r = 14 cm and h = 48 cm. Step 1: $l = \sqrt{14^2 + 48^2} = \sqrt{196 + 2304} = \sqrt{2500} = 50$ cm. Step 2: Lateral SA = $\pi \times 14 \times 50 = 700\pi \approx 2\,199$ cm². Step 3: Base area = $\pi \times 14^2 = 196\pi \approx 616$ cm². Step 4: TSA ≈ 2 815 cm². Teacher anchor: record this cone bucket value on the class Driving Question Board — it will be compared with pyramid and frustum buckets of equal volume in Lesson 8.
Lesson 2 **Surface Area of Pyramids (Square and Rectangular Base)**	Students observe photos of the grain storage pyramidal structures used in Kitale and Eldoret (or the roof trusses on traditional Kikuyu homesteads) and estimate how much roofing material covers each triangular face. Teacher distributes square-based and rectangular-based pyramid nets (pre-cut from manila paper) and asks students to identify and label each face before any formula is introduced. Key	For a square-based pyramid with base side l and slant height s: TSA = $l^2 + 4 \times (\frac{1}{2} \times l \times s) = l^2 + 2ls$. For a rectangular-based pyramid with base dimensions l × w and respective slant heights s_1 (over width faces) and s_2 (over length faces): TSA = $lw + 2 \times (\frac{1}{2}ls_1) + 2 \times (\frac{1}{2}ws_2) = lw + ls_1 + ws_2$. Pedagogical note — 'slant height' of a pyramid face is the perpendicular height of that triangular face measured from the base	Worked example 1 (square base): A maize storage shed roof in Kitale is a square pyramid with base side 6 m and slant height 5 m. TSA = $6^2 + 2(6)(5) = 36 + 60 = 96$ m². Worked example 2 (rectangular base): A rectangular pyramid roof, base 8 m × 5 m, slant heights 4.5 m (over 8 m faces) and 6 m (over 5 m faces). TSA = $(8\times5) + (8\times4.5) + (5\times6) = 40 + 36 + 30 = 106$ m². Teacher should cold-call at least three students to

	observation to elicit: a square-based pyramid has 1 square base + 4 identical isoceles triangles; a rectangular-based pyramid has 1 rectangular base + 2 pairs of non-identical triangles.	edge to the apex, NOT the edge length. Draw a clear diagram showing this distinction every time.	identify which slant height belongs to which pair of faces before computing.
Lesson 3 **Surface Area of a Sphere**	Teacher presents a football (as used in the Kenyan Premier League) and asks: if we were to cover this ball entirely with leather panels, how much leather would we need? Students first estimate by wrapping a ball with strips of paper/tape and measuring the paper used. Teacher then demonstrates the classical orange-peel proof: peel an orange, flatten the peel, and show it fills almost exactly 4 circles of the same radius as the orange. This provides intuitive evidence that SA = $4\pi r^2$ before any algebraic derivation.	The surface area of a complete sphere of radius r is SA = $4\pi r^2$. This equals exactly 4 times the area of the sphere's great circle (πr^2). A closed hemisphere (flat face sealed) has TSA = $2\pi r^2 + \pi r^2 = 3\pi r^2$, where $2\pi r^2$ is the curved dome and πr^2 is the flat circular base. Pedagogical note — students often confuse surface area of a sphere with volume ($4/3\ \pi r^3$). A useful memory anchor: surface area has r^2 (two-dimensional) and volume has r^3 (three-dimensional). Reinforce this distinction explicitly.	Worked example: A spherical water tank at a Mombasa hotel has diameter 3 m (r = 1.5 m). (a) How much steel sheet is needed to construct it? SA = $4\pi(1.5)^2 = 4\pi(2.25) = 9\pi \approx 28.27$ m². (b) If only the curved dome is constructed (open base hemisphere), Curved SA = $2\pi(1.5)^2 = 4.5\pi \approx 14.14$ m². Teacher should ask students to explain in words why the hemisphere dome uses exactly half the steel of the full sphere — wait 30 seconds before cold-calling two students.
Lesson 4 **Surface Area of a Hemisphere**	Students observe a dome-shaped sufuria lid and a half-cut avocado (common in Kenyan markets) and discuss which surfaces are exposed. Teacher places a hemisphere model flat-side down on a table and asks: how many distinct surfaces does this solid have? Students must identify and separately label (i) the curved dome surface and (ii) the flat circular face. The teacher then asks students to predict whether the curved surface or the flat face has greater area — most predict the flat face, which is incorrect for $r > 0$.	Curved Surface Area of a hemisphere = $2\pi r^2$ (exactly half of the full sphere's $4\pi r^2$). Flat circular base area = πr^2. Total Surface Area of a closed hemisphere = $2\pi r^2 + \pi r^2 = 3\pi r^2$. For an open hemisphere (bowl, dome with no base): TSA = $2\pi r^2$ only. Pedagogical note — always ask students to state whether the hemisphere is 'open' or 'closed' before they begin calculating, as this determines which formula applies. This is a common exam error source.	Worked example 1 (closed): A dome tent used by Maasai Mara tourists has a hemispherical shape with r = 2.1 m. Total canvas needed (closed): TSA = $3\pi(2.1)^2 = 3\pi(4.41) = 13.23\pi \approx 41.57$ m². Worked example 2 (open bowl): A decorative stone bowl of r = 0.35 m. Curved SA = $2\pi(0.35)^2 = 2\pi(0.1225) = 0.245\pi \approx 0.770$ m². Teacher connects to DQB: record that a hemispherical container's curved area = $2\pi r^2$ and note this will be compared to conical and frustum containers later.
Lesson 5 **Introduction to Frustums: Nets and Vocabulary**	Teacher holds up a typical Kenyan plastic bucket (truncated cone / frustum shape) and a tapered drinking cup. Students observe that neither is a complete cone — the top has been 'cut off.' Teacher asks: where did the missing top cone go? Students then unfold a paper frustum net (pre-made) and identify two circles (top and bottom) and the curved lateral strip. Key vocabulary to establish: frustum, large base radius R, small top radius r, perpendicular height h, slant height l. Teacher labels all five quantities on the board diagram and requires students to copy and label on their own nets.	A frustum of a cone is the solid remaining after a small cone is removed from the top of a larger cone by a cut parallel to the base. The curved surface area of a frustum = $\pi(R + r)l$, where R = large base radius, r = small top radius, and l = slant height of the frustum (NOT the full cone). The slant height is calculated as $l = \sqrt{h^2 + (R - r)^2}$. Pedagogical note — Lesson 5 focuses on vocabulary and net recognition only; full TSA computation is developed in Lesson 6. Do not rush into closed/open TSA in this lesson.	Guided discovery: Given a frustum with R = 10 cm, r = 4 cm, h = 9 cm. Step 1: $l = \sqrt{9^2 + (10-4)^2} = \sqrt{81 + 36} = \sqrt{117} \approx 10.82$ cm. Step 2: Curved SA = $\pi(10 + 4)(10.82) = \pi(14)(10.82) = 151.48\pi \approx 475.9$ cm². Teacher models this step-by-step, annotating each number on the diagram. Students then compute a second example (R = 7 cm, r = 3 cm, h = 8 cm) independently and compare with a partner before sharing out — teacher uses cold-call for two pairs.

Lesson 6 **Total Surface Area of Frustums of Cones — Open and Closed**	Students compare an open metal bucket (no lid, no base visible — open frustum) with a closed tin can shaped as a frustum (both ends sealed). Teacher asks: what additional surfaces exist on the closed version? Students must identify the two circular ends and state their areas in terms of R and r. Teacher then poses: if we wanted to manufacture both containers from sheet metal, which requires more material and by how much? Students write predictions before calculating.	Curved Surface Area of a frustum of a cone $= \pi(R + r)l$, where $l = \sqrt{h^2 + (R - r)^2}$. Open frustum (no bases, like an open bucket): TSA $= \pi(R + r)l$. Closed frustum (both circular ends sealed): TSA $= \pi(R + r)l + \pi R^2 + \pi r^2$. If only the large base is closed (most common bucket scenario): TSA $= \pi(R + r)l + \pi R^2$. Pedagogical note — teachers should require students to sketch and label the frustum before writing any formula, and to explicitly state 'open' or 'closed' as part of their working. Marks are typically awarded for correct identification in KNEC exams.	Worked example (Kenyan context): A metal water bucket used on a Nakuru dairy farm has R = 18 cm (bottom), r = 12 cm (top), and h = 30 cm. The bucket is open at the top but has a metal base. Step 1: $l = \sqrt{30^2 + (18-12)^2} = \sqrt{900 + 36} = \sqrt{936} \approx 30.59$ cm. Step 2: Curved SA $= \pi(18+12)(30.59) = \pi(30)(30.59) = 917.7\pi \approx 2\,882$ cm². Step 3: Base area $= \pi(18)^2 = 324\pi \approx 1\,018$ cm². Step 4: TSA ≈ 3 900 cm². Record on DQB alongside the cone bucket from Lesson 1 for comparison in Lesson 8.
Lesson 7 **Surface Area of Frustums of Pyramids**	Students observe photographs of a truncated pyramid — teacher can use images of graduated grain silos or the stepped base sections of traditional granaries in western Kenya (Kisii or Luhya architecture). The frustum of a pyramid has two polygonal bases (square or rectangular) connected by trapezoidal lateral faces. Teacher distributes a pre-cut net of a square pyramidal frustum and students identify: 2 square bases (different sizes) and 4 congruent trapezoids. Students measure the trapezoid dimensions on the net before any formula is given.	For a closed frustum of a square-based pyramid with top side a, bottom side b, and slant height l (perpendicular height of each trapezoidal face): TSA $= a^2 + b^2 + 2(a + b)l$. The slant height l of the trapezoidal face is found using: $l = \sqrt{h^2 + ((b-a)/2)^2}$, where h is the perpendicular height of the frustum. For a rectangular-based pyramidal frustum with top dimensions a × c and bottom dimensions b × d: TSA $= ac + bd + (a+b)l_1 + (c+d)l_2$, where l_1 and l_2 are slant heights over respective pairs of faces. Pedagogical note — the formula for square frustum is the most commonly tested; ensure students can derive it from 'two squares + four trapezoids' rather than memorising it in isolation.	Worked example (square frustum): A decorative stone planter in a Nairobi hotel lobby is a frustum of a square pyramid with top side a = 20 cm, bottom side b = 32 cm, and height h = 15 cm. Step 1: $l = \sqrt{15^2 + ((32-20)/2)^2} = \sqrt{225 + 36} = \sqrt{261} \approx 16.16$ cm. Step 2: TSA $= 20^2 + 32^2 + 2(20+32)(16.16) = 400 + 1024 + 2(52)(16.16) = 400 + 1024 + 1\,680.6 \approx 3\,104.6$ cm². Teacher cold-calls three students to explain which term in the formula represents which face type. Add result to DQB for Lesson 8 comparison.
Lesson 8 **Consolidation, Applications and Model Revision: Comparing All Three Buckets and Finalising the Surface Area Model**	Teacher displays three containers side by side (or as labelled diagrams): the cone bucket from Lesson 1, the frustum bucket from Lesson 6, and a hypothetical cylindrical bucket — all holding the same volume (e.g., 10 litres). Students observe and record all three surface area values computed across previous lessons on the DQB. Teacher asks: which shape uses the least material? which uses the most? Why might a manufacturer choose a frustum even if it requires more material than a cone? Students discuss in pairs for 90 seconds before class share-out.	Equal volumes do NOT guarantee equal surface areas — the shape of a solid independently determines how much material is needed to construct or cover it. Among common container shapes of the same volume: (i) a sphere has the minimum possible surface area (most efficient); (ii) a frustum uses more material than a cone of the same volume but is more stable and stackable; (iii) designers make trade-offs between material cost, stability, and manufacturability. Summary of all formulae covered: Cone TSA $= \pi rl + \pi r^2$; Square pyramid TSA $= l^2 + 2ls$; Sphere SA $= 4\pi r^2$; Hemisphere TSA $= 3\pi r^2$ (closed); Frustum of cone TSA $= \pi(R+r)l + \pi R^2 + \pi r^2$; Square	Final application task: A Kisumu factory manufactures three styles of 10-litre grain containers — conical, frustum (cone), and square pyramidal frustum. Students compute the TSA of each using given dimensions, determine which style uses least sheet metal, and write a recommendation memo to the factory manager justifying their choice mathematically. Teacher facilitates class discussion: 'Is cheapest always best?' connecting to real-world design values such as stackability, pour-ability, and aesthetics. DQB is fully resolved: students write a final answer to the driving question in their own words as an exit ticket.

		frustum TSA = $a^2 + b^2 + 2(a+b)l$. Pedagogical note — distribute the completed formula reference card at this point and require students to annotate each formula with a sketch showing what r, l, h, R, a, b represent.	